Stress and Lifestyle: How Students' Habits Influence Mental Well-being

A statistical analysis of lifestyle habits, stress levels, and academic performance

Abstract. This report investigates the relationship between students' daily habits and psychological well-being. Using a Kaggle dataset of 2,000 university students, the analysis studies how study time, sleep, physical activity, social habits, extracurricular activity, and GPA relate to stress levels. The workflow combines exploratory data analysis, PCA, outlier treatment, clustering, classification, and regression-based segmentation to identify the lifestyle patterns most associated with student stress and academic performance.

Data and research objective

This analysis investigates the relationship between students' daily habits and their psychological well-being. In particular, it explores how everyday lifestyle factors influence stress levels and contributes to a broader understanding of student well-being.

The dataset used in this project is the Student Lifestyle Dataset from Kaggle. It contains observations on 2,000 university students, including study hours, sleep, physical activity, social activity, extracurricular activity, GPA, and stress level.

The reference variable is Stress Level, divided into three classes: Low, Moderate, and High. The high-stress category is the most represented group, followed by moderate stress, while low stress is the least frequent category.

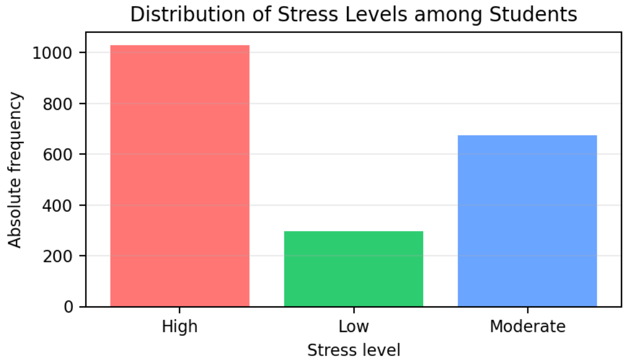

Figure 1 - Absolute frequency distribution of stress levels.

Which factors have the strongest effect on students' stress levels?

A Principal Component Analysis (PCA) was initially used to identify the variables that explain the largest share of variation in the dataset. The first four components explain approximately 82% of the total variance. This suggests that a reduced set of variables can still retain most of the relevant information.

The variables that appear most informative in the initial analysis are study hours, physical activity hours, social hours, and extracurricular hours. Study hours and physical activity hours are especially relevant, since they explain a large part of the variability and are strongly related to stress levels.

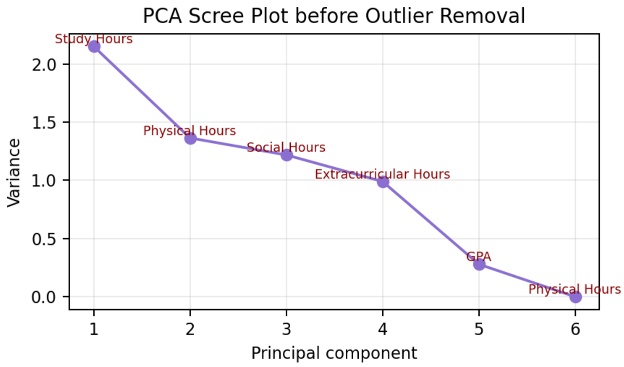

Figure 2 - PCA scree plot before outlier removal.

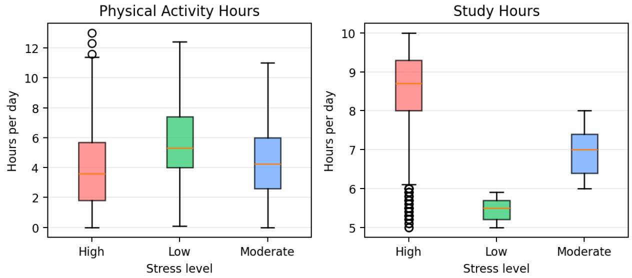

Figure 3 - Boxplots before outlier removal.

The exploratory plots show a clear relationship between study time and stress: students who study more tend to experience higher stress, while students with lower study hours are more often associated with lower stress. Physical activity appears to work in the opposite direction, since less stressed students generally dedicate more time to exercise.

Some observations in the high-stress group appear inconsistent with the expected pattern, since they present values more similar to low-stress students. These points were treated as outliers using the Interquartile Range (IQR) criterion. In total, 79 observations were removed to reduce distortion in the analysis.

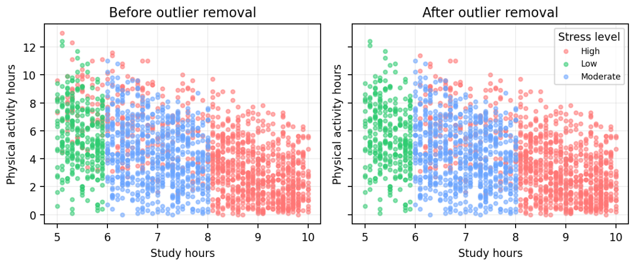

Figure 4 - Scatter plots before and after outlier removal.

How does the analysis change after removing outliers?

Because PCA is sensitive to extreme values, the analysis was repeated after removing anomalous observations. After cleaning the data, the most informative variables remain study hours, sleep hours, extracurricular hours, and physical activity hours. The first components explain about 79% of the variance, which is acceptable for dimensionality reduction.

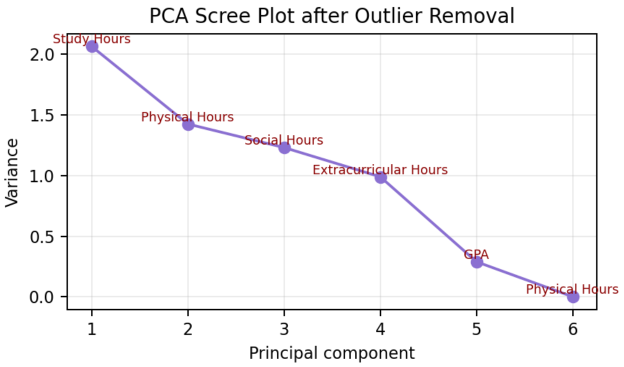

Figure 5 - PCA scree plot after outlier removal.

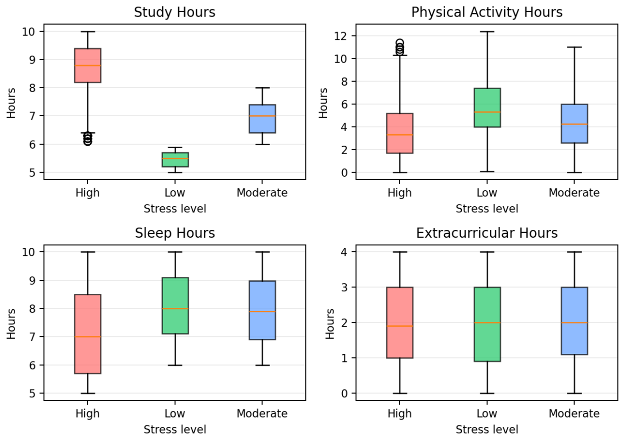

Figure 6 - Stress-level boxplots after cleaning.

After cleaning, differences across stress levels become clearer. Study hours remain the variable with the largest dispersion across the three stress categories. Physical activity and sleep decrease as stress increases, suggesting that heavier academic pressure may reduce the time dedicated to rest and exercise.

How are the variables correlated?

The correlation matrix shows that study hours, extracurricular hours, sleep hours, and physical activity hours are not strongly correlated with one another. This means that each variable captures a different aspect of student lifestyle.

The strongest relationships are negative: students who study more tend to spend less time on physical activity, and students who spend more time on physical activity may sleep less. This reflects time-allocation trade-offs, where dedicating more time to one activity often means reducing time for another.

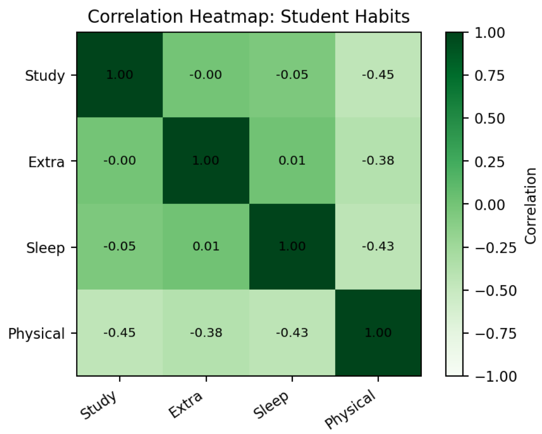

Figure 7 - Correlation heatmap of lifestyle variables.

Can statistical clustering replace a psychologically based classification?

The comparison between the original stress-level classification and the clustering obtained through a model-based method highlights important differences. The original labels separate students into three psychologically meaningful groups: high stress, low stress, and moderate stress. The clustering approach, instead, ignores the original labels and groups observations only according to statistical patterns.

The resulting clusters are more balanced in size than the original classes, but they do not perfectly match the psychological interpretation of stress. This suggests that clustering can reveal useful structure in the data, but it should not replace classifications based on domain knowledge and psychological research.

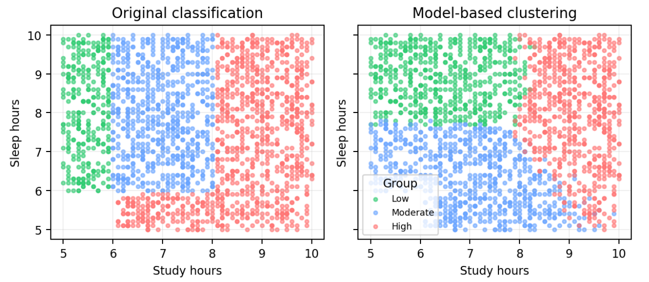

Figure 8 - Original classification versus model-based clustering.

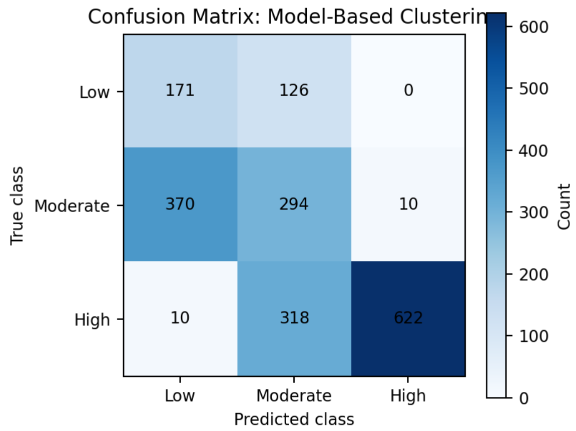

Figure 9 - Confusion matrix for model-based clustering.

The unsupervised model has difficulty distinguishing the moderate-stress category, with many observations assigned to either low or high stress. This confirms that the middle category is less sharply separated in the data.

Can we predict a student's stress level from lifestyle habits?

To evaluate predictive performance, the dataset was split into a training set and a test set. Several Gaussian classification models were compared using cross-validation and BIC. These criteria help evaluate prediction error while penalizing excessive model complexity, reducing the risk of overfitting.

After selecting the best model, predictions were evaluated on the test set. The confusion matrix shows very strong performance, with an overall accuracy of 90.91%. The first class is classified almost perfectly, while the second and third classes show a small degree of overlap.

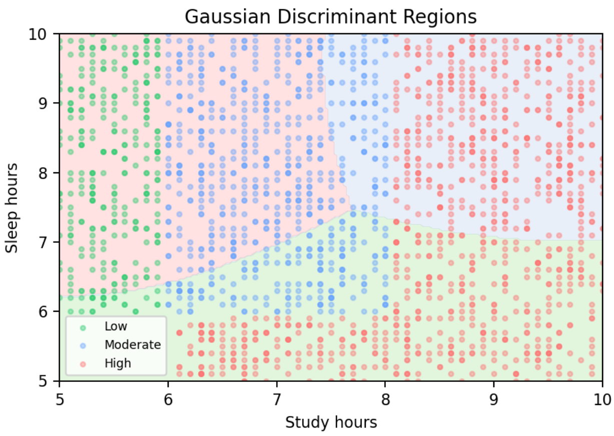

Figure 10 - Gaussian discriminant regions.

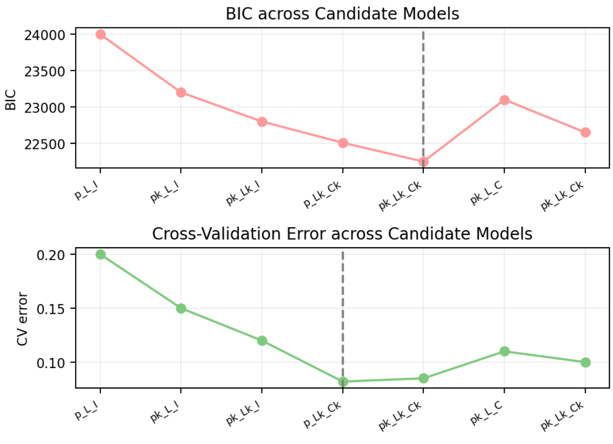

Figure 11 - Model-selection criteria.

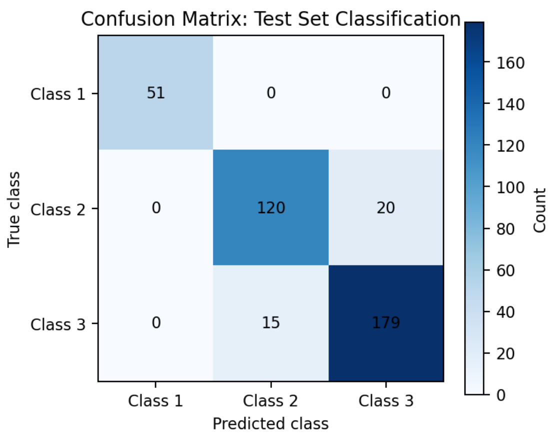

Figure 12 - Confusion matrix for test-set classification.

Stress and academic performance: regression and clustering

The analysis then investigates the relationship between stress and academic performance, measured by GPA. A regression-based clustering approach is used to assess stress both as a predictor and as a clustering variable.

The correlation heatmap shows that GPA has a strong positive correlation with study hours, suggesting that greater study effort is associated with better academic performance. GPA also has a moderate negative correlation with physical activity hours, while the other variables show much weaker relationships.

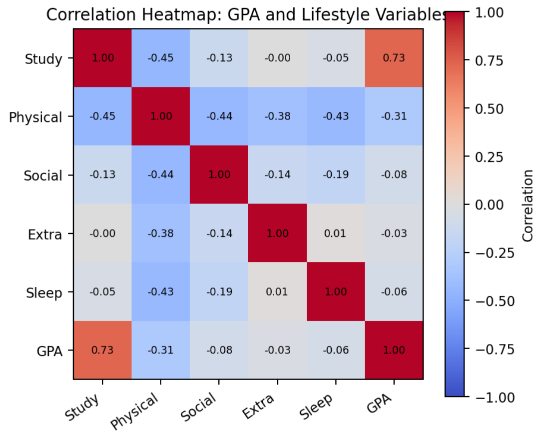

Figure 13 - Correlation heatmap: GPA and lifestyle variables.

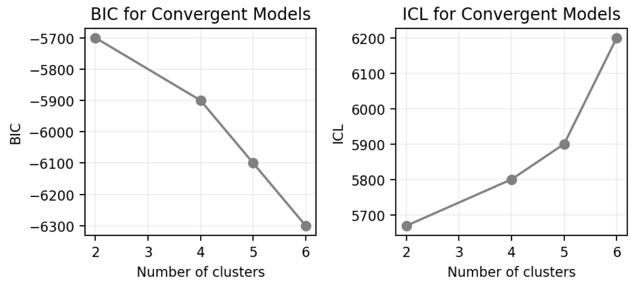

Figure 14 - Convergent model indices.

For student segmentation, a two-cluster model is more stable than a three-cluster solution, which does not converge reliably. The optimal segmentation distinguishes two groups: Cluster 1 is mainly composed of students with moderate or low stress, while Cluster 2 is dominated by students with high stress.

Figure 15 - Distribution of stress levels across clusters.

Do study hours and sport influence academic performance within clusters?

The relationship between GPA and study hours confirms the correlation results. Increasing study time is associated with a higher GPA in both clusters, although students in Cluster 1 tend to achieve higher results for the same amount of study time.

Physical activity is generally associated with a slight decrease in GPA. This effect is more visible in Cluster 1, while students in Cluster 2 appear less affected by this variable.

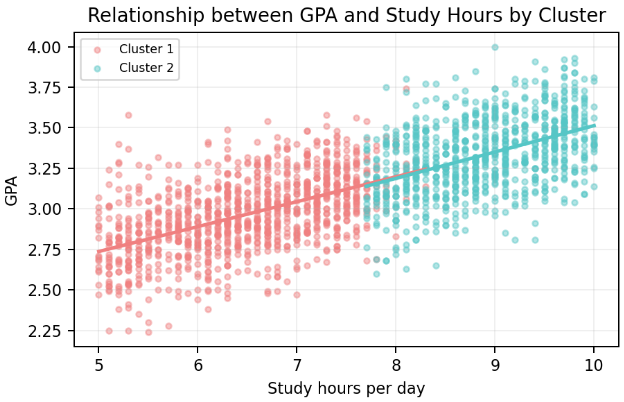

Figure 16 - Relationship between GPA and study hours by cluster.

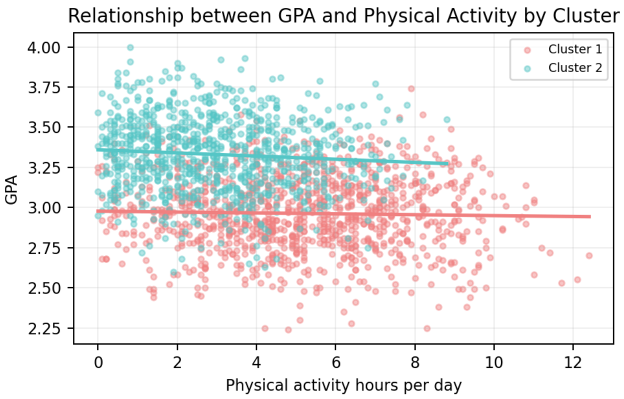

Figure 17 - Relationship between GPA and physical activity by cluster.

Is GPA influenced by stress level?

The effect of stress on predicted GPA differs between the two clusters. In Cluster 1, academic performance is relatively stable across stress levels, with a slight decrease under high stress. In Cluster 2, GPA is more sensitive to stress variation: students with moderate stress achieve the best average outcomes, while students with low stress show the lowest predicted GPA.

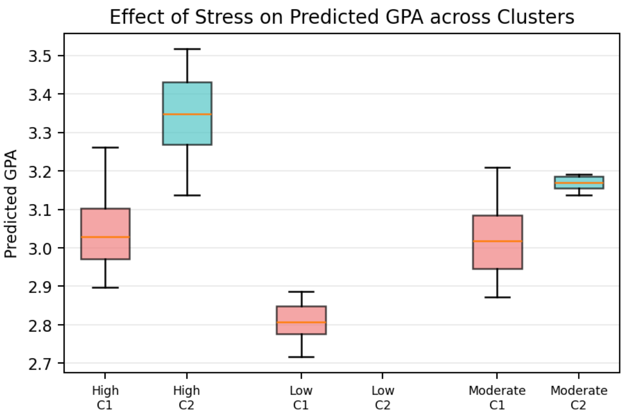

Figure 18 - Effect of stress on predicted GPA across clusters.

How can these results improve student support?

Using stress both as a predictor in regression and as a clustering variable makes it possible to capture two dimensions of the phenomenon: its direct relationship with academic performance and its ability to distinguish groups of students with different behavioral patterns.

The results suggest that stress does not affect all students in the same way. For some students, a moderate amount of pressure may be associated with high performance, while for others stress may represent a risk factor. This highlights the need for targeted support strategies based on students' sensitivity to stress, rather than a single generalized approach.

Conclusion. Lifestyle variables, especially study time, physical activity, and sleep, are meaningfully associated with stress and academic performance. The results support targeted student-support strategies rather than a single generalized intervention.